

HARMAN PAPNEJA

AI Software Engineer | Agentic Systems & Backend AI Infrastructure

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PROFESSIONAL SUMMARY

AI Software Engineer with 2+ years of experience building production Agentic AI systems using Python, LLMs, LangChain/LangGraph, FastAPI, and Retrieval-Augmented Generation (RAG). Skilled in designing tool-use logic that lets LLMs interact with PostgreSQL/pgvector-backed databases and internal APIs, enforcing strict structured output via Pydantic to keep autonomous agents schema-safe. Experienced with asynchronous Python (Asyncio), REST APIs (Flask/FastAPI), MongoDB, and end-to-end LLM evaluation pipelines achieving 90%+ accuracy against human benchmarks, while scaling AI backends to 2.2M+ records annually at 99.9% uptime.

TECHNICAL SKILLS

Programming Languages: Python (Asyncio, Pydantic), SQL (PostgreSQL, MySQL), NoSQL (MongoDB)

Generative AI / LLM: Large Language Models (LLM), OpenAI, Groq, Google GenAI SDK, Prompt Engineering, Chain-of-Thought (CoT) Prompting, Retrieval-Augmented Generation (RAG), Structured Output (Pydantic), Vision-Language Models (VLM), Context-Window Management, Context Caching, LLM Guardrails, Semantic Analysis

Agentic AI & Orchestration: LangChain, LangChain Expression Language (LCEL), LangGraph, Agentic Workflows, Autonomous Agents, Function/Tool Calling, Agentic Routing, Recommendation Systems

ML / Data & Vector Databases: TensorFlow, Scikit-learn, PyTorch (working knowledge), Pandas, NumPy, HuggingFace, pgvector, Pinecone, ChromaDB

Backend & Infrastructure: FastAPI, Flask, REST APIs, PostgreSQL, JSON-Schema Validation, MongoDB, Event-Driven Architecture, Background Jobs, Tavily Search API, Playwright, Asynchronous Processing

PROFESSIONAL EXPERIENCE

AI Engineer, CSsoft Solutions

Aug 2026 – Present

Astravi — AI-Driven Employee-Work Intelligence & Recommendation Platform

- Developing Astravi, an AI-driven employee-work intelligence platform that converts minute-level activity signals and VLM-based screenshot captions into semantic task summaries and privacy-safe productivity recommendations, using Python and LangChain.
- Designed a resilient screenshot-captioning pipeline with unique screenshot identities, MongoDB checkpointing, and retry/resume logic, enabling fault-tolerant processing of high-volume activity data.
- Built a decision AI agent that governs the recommendation flow, determining when a real opportunity exists before triggering generation and suppressing false signals from idle or self-generated UI states.
- Engineered a privacy-by-design recommendation architecture (current_work, TaskSummary, RecommendationDecision) integrated with Search APIs, stripping sensitive values, URLs, and identifiers before LLM-based generation.

AI Software Engineer, Altruist Technologies

May 2024 – Aug 2026

SENSE AI — Call Audit & Sentiment Automation

- Architected a high-throughput Generative AI backend in Python, scaling call-audit automation to 2.2M+ calls/year at 99.9% uptime.
- Cut API token costs and latency via a custom context-caching layer, persisting structured LLM evaluation logs in MongoDB.
- Improved automated QA alignment to 90%+ vs. human benchmarks using dynamic prompt engineering (CoT + Gamma-style constraints) to evaluate intent and behavior via LLMs.
- Built structured AI coaching pipelines with strict Pydantic/JSON-Schema outputs, autonomously surfacing conversational gaps and soft-skill coaching recommendations.
- Cut false-positive compliance flags by 80% via logic-verification constraints in the model workflow, informing recovery strategies for clients including Bank of Baroda, Razorpay, and Magnificat.

SENSE AI — Agentic Email Orchestration (LangGraph)

- Engineered an autonomous agentic AI workflow using LangGraph and Python/Flask to orchestrate multi-turn email context (up to 5 threads), generating next steps, replies, and summaries with <3s latency.
- Implemented agentic routing and tool calling via the Tavily Search API, letting the agent verify drafted facts against real-time web data and prevent LLM hallucination.
- Achieved 100% data-extraction accuracy on missing draft fields via dynamic, few-shot Gamma prompts with strict JSON-Schema validation for frontend integration.
- Optimized backend throughput via asynchronous Python (Asyncio) processing and thread-truncation, managing the LLM context window to cut per-request cost and latency.

PERSONAL PROJECTS

Agentic Job Applier Python, LangChain (LCEL), LangGraph, FastAPI, Llama 3.3 (Groq), Playwright, ChromaDB, HuggingFace

- Built an end-to-end autonomous AI agent using LangChain Expression Language (LCEL) to source, evaluate, and apply to job roles with full autonomy.
- Architected a dual-layer RAG “Career Brain” using ChromaDB (vector database) and HuggingFace embeddings, with sub-second inference via Llama 3.3 (70B) on Groq for real-time JD-to-resume matching.
- Developed a Playwright-based web-automation layer with cookie injection to bypass MFA/bot-detection, plus state-aware verification for complex DOM structures.
- Implemented a self-learning memory layer that persists novel form data (notice period, CTC, etc.) and normalizes URLs, enabling 0% manual re-entry and eliminating duplicate processing.

EDUCATION & CERTIFICATIONS

Bachelor of Computer Science and Engineering — Chitkara University

2020 – 2024

CGPA: 9.48 / 10

Certifications: PG Program in Data Science & Analytics – Imarticus Learning; Advanced Data Analytics – Google; Data Science with Python – Simplilearn.